مدينة زويل للعلوم والتكنولوجيا
Zewail City of Science and Technology

CSAI 498 / CSAI 499 – Graduation Project Proposal

Project Title:

Smart Pre-Recycling: Online/Offline AI-based Waste Classifier

Team Members:

Name	ID	Program
Sara Ehab	202202146	Data Science and Artificial Intelligence (DSAI)
Salma Ramy	202201759	Data Science and Artificial Intelligence (DSAI)
Shahd Tarek	202202263	Data Science and Artificial Intelligence (DSAI)
Sandy Mohamed	202202034	Software Development

Supervisor:

Dr. Mohamed Maher Ata
Associate Professor, School of Computational Sciences and Artificial Intelligence, Zewail City

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Abstract:

The Smart Pre-Recycling project presents a smart automated approach to waste management, addressing the inefficiencies and safety concerns associated with manual waste sorting. Traditional sorting methods are labor-intensive, error-prone, and expose workers to hazardous materials, resulting in contamination of recycling streams and harmful environmental outcomes. These limitations highlight the urgent need for a safer and more efficient pre-recycling process that ensures accurate material classification with minimal human intervention.

This project proposes an AI-driven waste classification system that integrates computer vision and deep learning techniques to automatically identify and categorize waste materials efficiently and accurately. The system utilizes a conveyor-based mechanism to transport items, while servo motors dynamically redirect each object toward its designated collection path according to the classification results produced by the trained model. A complementary mobile application enables real-time monitoring, performance visualization, and environmental impact assessment.

The proposed prototype aims to achieve a classification accuracy exceeding 90%, with at least an 80% reduction in manual involvement, therefore enhancing both operational safety and sorting efficiency. By combining automation, artificial intelligence, and data analytics, Smart Pre-Recycling contributes to the development of sustainable, scalable, and intelligent waste management systems that advance environmental conservation.

Problem Statement and Motivation

The increase of solid wastes has become a critical global challenge, placing immense pressure on recycling facilities and environmental management systems. A key bottleneck in the recycling process is the manual sorting of waste materials, which remains the most effective, labor-intensive, hazardous, and error-prone stage of the recycling process as the workers performing manual sorting are frequently exposed to sharp, toxic, or infectious materials, posing significant health and safety risks. Moreover, human error in classification often leads to contamination of recyclable streams, reducing material recovery efficiency and increasing operational costs.

The problem extends beyond workers to affect municipal authorities, recycling industries, and the environment at large since the ineffective waste sorting decreases the economic value of recyclable materials, contributes to land and water pollution, and undermines sustainability

efforts. Despite technological advancements in recycling, automated pre-sorting systems remain limited in developing regions due to high costs and lack of adaptable AI-based solutions.

Addressing this challenge is essential to promoting safer, more efficient, and environmentally responsible recycling operations. By developing an AI-driven automated classification system capable of accurately sorting waste with minimal human involvement, this project seeks to improve recycling accuracy, enhance worker safety, and contribute to a more sustainable circular economy.

Proposed Solution

The Smart Pre-Recycling system introduces an intelligent and automated approach to the most critical and hazardous phase of the recycling process which is the waste sorting. The core idea is to utilize artificial intelligence, computer vision, and mechatronic control to classify and direct waste materials efficiently, safely, and with minimal human intervention.

Unlike conventional waste sorting systems that rely on fully manual labors, the proposed solution is cost-effective, adaptable, and capable of operating in both online (cloud-based) and offline (on-device) modes, ensuring accessibility across various environments, including developing regions with limited connectivity.

The system comprises three main components:

1. **AI Core:** A deep learning model trained on a comprehensive waste image dataset to accurately classify materials into categories such as plastic, glass, metal, paper, and organic waste.

2. **Conveyor-Based Hardware Integration:** A treadmill-like conveyor belt carries waste items under a camera for image capture and classification. Based on the AI output, servo motors dynamically direct each item toward its appropriate collection path, replacing the need for complex robotic arms.

3. **Mobile Application Interface:** A user-friendly mobile application that displays real-time classification results, sorting statistics, environmental impact data, and system performance metrics.

This integrated approach leverages AI automation, mechanical precision, and real-time monitoring, making Smart Pre-Recycling an innovative and scalable solution for safer, faster, and more sustainable waste management.

Project Scope

In Scope

Must-Have Features

- **AI-Based Waste Classification:** Development and training of a deep learning model capable of accurately identifying and classifying waste into predefined categories such as plastic, glass, metal, paper, … etc.

- **Conveyor and Servo Mechanism:** Implementation of a conveyor-based hardware system that transports waste items under a camera (ESP32-CAM) for image capture, with servo motors directing each item to its designated category based on the AI model's decision.

- **Real-Time Data Acquisition:** Integration of a camera module using ESP32-CAM which is a microcontroller with an attached camera to capture live images and feed them into the AI model for immediate classification.

- **Mobile Application Interface:** Design of a mobile application to display classification results, system statistics, and environmental impact metrics in real time.

- **Dual Operation Modes:** Support for both online (cloud-based) and offline (on-device) operation to ensure flexibility and accessibility in different environments.

Could-Have Features

- **User Leaderboard or Community Dashboard** to encourage eco-friendly behavior by tracking and comparing users' recycling contributions.

- **Automated Dataset Expansion** using self-collected images from the camera for continuous model improvement.

- **Voice or Notification Alerts** to inform users about classification results or errors in real time.

Out of Scope

- **Robotic Arm Integration:** The system will not include robotic arms or advanced robotic manipulators.
- **Full Recycling Process:** Processing, compression, or remanufacturing of materials after sorting is beyond the project's current scope.
- **Complex Hazardous Waste Handling:** The project will not manage chemical or biomedical waste requiring specialized treatment.

Assumptions & Limitations

- The AI model's accuracy depends on the quality and diversity of the training dataset used.
- The hardware setup (camera, conveyor, servo motors) is assumed to function reliably under controlled lab conditions.
- Environmental variables such as lighting, background noise, or camera positioning may affect real-time classification accuracy.
- The mobile application will primarily focus on monitoring and visualization, not on controlling hardware remotely.
- System latency may vary slightly depending on whether the model runs locally (offline) or via cloud inference (online).

High-level timeline

Phase	Description	Duration (weeks)	Deliverables
Research & Requirement Analysis	Conduct a detailed literature review on waste classification technologies, collect and preprocess the dataset, and define technical and functional requirements for the AI model,	4	Dataset prepared, requirements, specification finalized.

	conveyor system, and mobile application.		
Design & Planning	Design the overall system architecture, including the AI model structure, hardware control flow, and app interface design. Select and justify the technology stack and integration methods.	3	System architecture diagrams, detailed design documentation
Implementation Part 1	Develop and train the deep learning model for waste classification. Begin integrating the AI model with the conveyor and servo control mechanisms. Implement the core features of the mobile app for data visualization.	10	Trained AI model, partially integrated hardware setup, and functional mobile app prototype
Implementation Part 2	Complete hardware–software integration. Optimize model performance, improve servo response accuracy, and finalize the mobile app interface for real-time system feedback and statistics.	6	Fully integrated prototype, refined AI model, and complete mobile app
Testing & Evaluation	Perform system testing and validation, evaluate model performance using metrics (accuracy, precision, recall, F1-score), and prepare final documentation and demonstration materials.	4	Tested prototype, performance evaluation report, and final documentation

Technology Stack & Theoretical Basis

Programming Languages, Frameworks, Tools, and APIs

- **Python:** Used as the primary programming language for AI model development and data processing due to its simplicity, wide adoption in the AI community, and availability of powerful machine learning libraries.

- **TensorFlow / PyTorch:** Employed to build and train deep learning models, specifically convolutional neural networks (CNNs), for accurate image-based waste classification. PyTorch allows flexibility during experimentation, while TensorFlow ensures scalability for deployment.

- **NumPy, Pandas, Scikit-learn:** Utilized for data preprocessing, analysis, and performance evaluation. These libraries provide essential utilities for handling datasets, computing metrics, and splitting training and testing data.

- **OpenCV and Albumentations:** Used for image acquisition, preprocessing, and augmentation to enhance dataset diversity and model robustness against lighting and environmental variations.

- **Raspberry Pi (Python) and Arduino (C/C++):** Implemented for hardware integration and real-time control. The Raspberry Pi handles image capture and inference, while Arduino operates the conveyor belt and servo motors that direct waste items based on AI classification outputs.

- **Flutter (Dart) or React Native (JavaScript/TypeScript):** Chosen for cross-platform mobile app development, enabling real-time visualization of classification results, statistics, and environmental impact.

- **Firebase / SQLite:** Used for data storage and synchronization. Firebase supports real-time cloud connectivity, while SQLite enables lightweight offline operation.

- **Git and GitHub:** Adopted for version control, collaborative development, and tracking project progress efficiently.

Theoretical Basis

- The system is grounded in computer vision and deep learning principles, focusing on Convolutional Neural Networks (CNNs) for visual waste classification.
- The model lifecycle includes data collection, preprocessing, training, validation, and deployment on edge hardware (Raspberry Pi).
- Real-time image data from the camera is processed by the CNN to classify materials such as plastic, glass, metal, paper, and organic waste.
- Classification results are sent to the hardware module, where servo motors guide each waste item toward its appropriate category.
- The mobile application serves as a monitoring interface that displays sorting statistics, accuracy, and environmental insights.

Program Focus: DSAI

This project aligns with the Data Science and Artificial Intelligence (DSAI) program by focusing on:

- Comprehensive data lifecycle management from collection to deployment.
- Design and optimization of AI models for real-time inference on embedded systems.
- Integration of machine learning with hardware for autonomous decision-making.
- Continuous model evaluation and improvement using standard performance metrics (accuracy, precision, recall, and F1-score).
- Contribution to sustainable AI applications that bridge data-driven intelligence with real-world environmental impact.

Success Metrics & Evaluation Plan

Success Metrics

- **Model Accuracy:**
  The AI waste classification model should achieve at least 90% accuracy in identifying and categorizing materials such as plastic, metal, glass, paper, and organic waste.

- **Performance and Latency:**
  The system should maintain an inference time ≤ 12 seconds per item (5-10 items per minute) ensuring smooth real-time operation.

- **Usability and User Satisfaction:**
  The mobile application should offer a clear, intuitive interface with update latency below 5 seconds, and positive user feedback gathered through surveys during prototype testing.

- **Reliability:**
  The conveyor–servo mechanism should consistently sort at least 95% of items without mechanical errors, demonstrating operational stability.

- **Data Quality and Model Robustness:**
  The trained model should generalize effectively across diverse waste types and lighting conditions, validated through cross-validation and confusion matrix analysis.

- **Risk Reduction:**
  The system should reduce human involvement in the sorting process by at least 80%, improving both safety and efficiency.

Evaluation Plan

- **Model Evaluation:**
  The AI model will be assessed using standard machine learning metrics (accuracy, precision, recall, and F1-score) to ensure balanced performance across all waste categories. Testing will be performed on both validation and unseen datasets to evaluate generalization.

- **Performance Testing:**
  Hardware latency will be measured to verify responsiveness and synchronization

between the AI model and mechanical system.

- **Reliability Testing:**
  Continuous sorting tests will be conducted under varying lighting and environmental conditions to evaluate system stability and robustness.

- **Usability Evaluation:**
  A small group of participants will interact with the mobile app prototype. Their feedback on clarity, responsiveness, and overall experience will be collected through structured surveys and used to refine the interface.

- **Comparative Benchmarking:**
  The developed system's classification accuracy and sorting efficiency will be compared with existing academic or open-source waste classification models to demonstrate improvement and innovation.

- **Documentation and Validation:**
  All testing results, performance metrics, and user feedback will be documented and validated by the project team to ensure transparency and repeatability.

Team Roles & Responsibilities

Team Member	Program	Primary Role	Technical Contribution
Sara Ehab	DSAI	Project Lead & AI Model Developer, and Hardware integration.	• Lead the design, training, and optimization of the deep learning model for waste classification. • Oversee dataset collection, preprocessing, and augmentation to ensure model robustness and accuracy. • Coordinate team progress and ensure project milestones are

			met according to the proposed timeline. ● Assist in hardware integration. ● Test and monitor the model's performance, accuracy, and speed during operation.
Salma Ramy	DSAI	AI Model Developer, and Hardware integration.	● Manage data preprocessing, cleaning, and balancing to maintain dataset quality. ● Conduct model evaluation using performance metrics such as accuracy, precision, recall, and F1-score. ● Test and monitor the model's performance, accuracy, and speed during operation. ● Assist in hardware integration.
Shahd Tarek	DSAI	AI Model Developer, and Hardware integration.	● Perform preprocessing operations such as data cleaning and class balancing to preserve dataset integrity and reliability. ● Test and monitor the model's performance, accuracy, and speed during operation.

			● Work on improving model efficiency and reducing response time for faster sorting. ● Help analyze system results and ensure the model performs well under different conditions. ● Assist in hardware integration.
Sandy Mohamed	SWD	Software developer and hardware integration	● Design and implement the mobile application interface using Flutter or React Native. ● Develop dashboards to visualize classification results, performance statistics, and environmental impact. ● Handle backend integration with Firebase/SQLite for real-time data storage and synchronization. ● Ensure responsive, secure, and user-friendly app design that complements AI model outputs. ● Assist in hardware integration.

Communication Plan

Meeting frequency:

- Weekly team meetings will be held to discuss progress, challenges, and upcoming tasks.

- Bi-weekly meetings with the project advisor will be scheduled to review milestones and receive feedback.
- Urgent meetings may be arranged when urgent technical or design issues arise.

Communication Tools:

- **Google meet/ Zoom:** for virtual meetings and advisor check-ins.
- **WhatsApp Group:** for daily communication, quick updates, and coordination among team members.
- **Email:** for formal correspondence and advisor communication.

Documentation and Collaboration Platforms:

- **Google Drive:** for storing project files, shared documents, datasets, and meeting notes.
- **GitHub:** for version control, code sharing, and collaborative development of the AI model and mobile application.

References

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